

Finite δS is Enclosure of Mixing Entropy. Only on the Ball that Encloses.

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Abstract

Paper 60: one enclosing ball has $\delta S = 2h(\alpha)$. Paper 62: 216 balls, δS tracks P_{flat} . This note asks whether they are the same fraction,

$$f_S = \delta S / S_{\text{global}} \stackrel{?}{=} f_E = P_{\text{flat}} / M_{\text{global}}.$$

$N = 12$, $R = 3$, $\alpha = 0.005, 0.02, 0.08$. $\rho(f_S, f_E) = 0.991, 0.996, 0.999$. RMS 0.098, 0.062, 0.028. On the centered ball both recover 1 to $< 0.5\%$. On the 120 source-inside balls, median $|f_S - 1| = 0.39, 0.34, 0.28$: those balls leak. Auditor $N = 10$: $\rho = 0.998$ at both α ; centered $f_S = 0.993, 0.996$. Correlation confirmed. Absolute $f = 1$ on every source-inside ball is refuted.

Finite δS is Shannon of the transfer times how much of the packet sits in the ball. Gravity remains $\sum \delta e$ plus inherited Gauss. Not Clausius.

Admission. *I pre-registered $f = 1$ on every ball that contains the source site. That gate failed. I did not move it. The centered ball recovers 1. The result is the correlation, not a first-law constant.*